Five number summary

1 For each of the the following sets of data:

i Sort the data in ascending order.

ii Find the maximum value.

iii Find the minimum value.

iv Find the median value.

v Find Q_1 for this data set.

vi Find Q_3 for this data set.

a The data set shows number of points scored by a basketball team in each game of their previous season:

$$75, 53, 84, 66, 89, 55, 63, 70, 92, 51, 90, 55, 81, 87, 68$$

b The data set shows marks in an end-of-year exam for a class of students:

$$59, 53, 75, 80, 82, 96, 81, 79, 64, 58, 77, 62, 62, 86$$

c The data set shows finishing times (in minutes) of the competitors in a 1500-metre swimming race:

$$24.41, 22.95, 21.88, 24.19, 16.12, 25.64, 16.83, 23.62, 24.52, 23.74, 19.44$$

2 There is a test to measure the Emotional Quotient (EQ) of an individual. Here are the EQ results for 21 people, listed in ascending order:

$$92, 94, 100, 103, 103, 105, 105, 109, 110, 113, 114,$$
$$114, 116, 118, 118, 119, 120, 125, 125, 126, 130$$

a Determine the median EQ score.

b Determine Q_1 for this data set.

c Determine Q_3 for this data set.

3 Consider the following set of scores:

$$10, 11, 12, 13, 15, 17, 19, 20$$

a Within what range do the middle 50% of scores lie?

b State another name for the middle 50% of scores.

4 If there are 78 scores in a set of data, in which position will the lower quartile lie?

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5 The table shows the luggage weight, in kilograms, of 30 passengers.

Weight	Frequency
16	5
17	5
18	2
19	4
20	6
21	4
22	4

a What is the mean check in weight? Round your answer to two decimal places.

b Determine the median, Q_1, and Q_3.

c In which quartile does the mean lie?

Box plots

6 Construct a box plot for each five number summary:

a

- Median $= 47$
- Lower Quartile $= 33$
- Upper Quartile $= 61$
- Lowest score $= 16$
- Highest score $= 71$

b

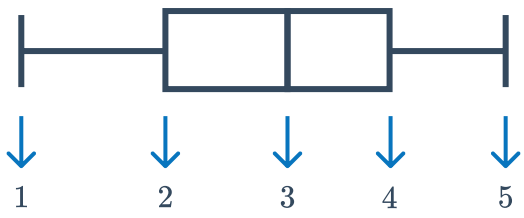
- Median $= 36$
- Lower Quartile $= 28$
- Upper Quartile $= 42$
- Lowest score $= 20$
- Highest score $= 52$

c

- Median $= 35$
- Lower Quartile $= 25$
- Upper Quartile $= 60$
- Lowest score $= 5$
- Highest score $= 75$

7 Consider the box plot and statistics listed below:

- Median $= 47$
- Lower Quartile $= 33$
- Upper Quartile $= 61$
- Lowest score $= 16$
- Highest score $= 71$

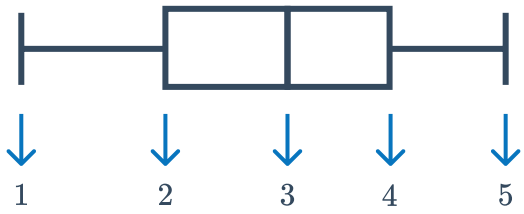


Which number should go in position 2 on the box plot?

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8 Consider the box plot and statistics listed below:

- Median $= 36$
- Lower Quartile $= 28$
- Upper Quartile $= 42$
- Lowest score $= 20$
- Highest score $= 52$



Which number should go in position 4 on the box plot?

9 Using the information in the table, create a box plot to represent this data set:

Minimum	5
Lower Quartile	25
Median	35
Upper Quartile	60
Maximum	75

10 A geography teacher has marked a set of tests. She wants to represent the results in a box plot. She has already sorted her data and created the table shown:
Create a box plot to match the data in the table.

Minimum	8
Lower Quartile	10
Median	16
Upper Quartile	24
Maximum	28

11 Consider the following data set:

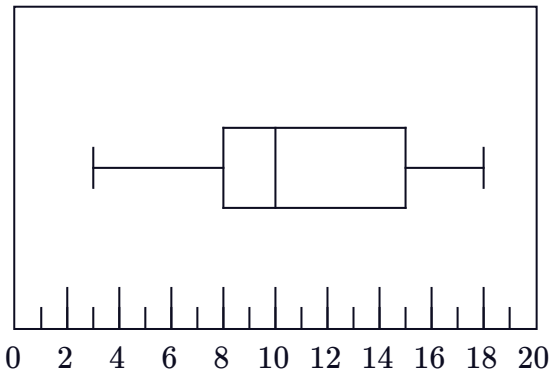
$$20, 36, 52, 56, 24, 16, 40, 4, 28$$

a Find the five number summary.

b Construct a box plot for the data.

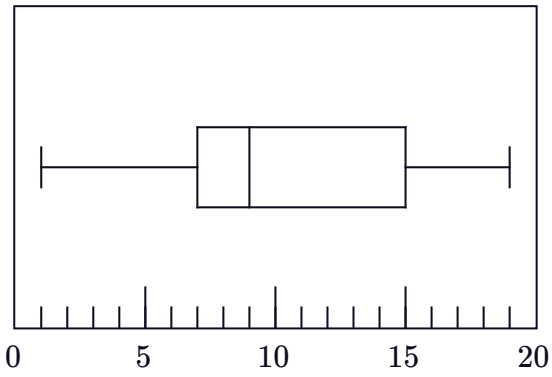
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12 For the box plot shown, find the following:



- **a** Lowest score
- **b** Highest score
- **c** Range
- **d** Median
- **e** Interquartile range

13 Consider the box plot shown:



- **a** State the percentage of scores that lie between each of the following values:
  - **i** 7 and 15
  - **ii** 1 and 7
  - **iii** 19 and 9
  - **iv** 7 and 19
  - **v** 1 and 15
- **b** In which quartile is the data the least spread out?

Applications

14 To gain a place in the main race of a car rally, teams must compete in a qualifying round. The median time in the qualifying round determines the cut off time to make it through to the main race. Below are some results from the qualifying round:

- 75% of teams finished in 159 minutes or less.
- 25% of teams finished in 132 minutes or less.
- 25% of teams finished between with a time between 132 and 142 minutes.

- **a** Find the median time for the qualifying round.
- **b** Hence, state the cut off time required in the qualifying round to make it through to the main race.
- **c** Find the interquartile range in the qualifying round.
- **d** In the qualifying round, the ground was wet, while in the main race, the ground was dry. To make the times more comparable, the finishing time of each team from the qualifying round is reduced by 5 minutes.
  Find the new median time from the qualifying round.

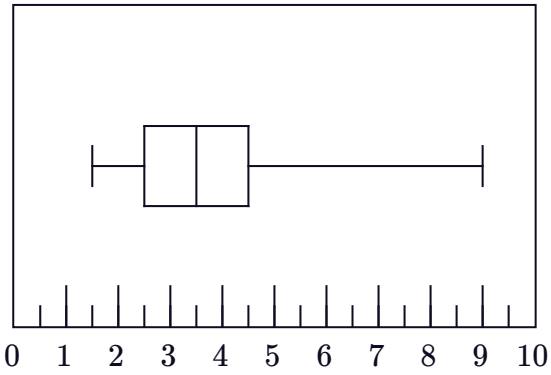
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15 In competition, a diver must complete 8 rounds of dives. Her scores for the first 7 rounds are:

$$7.3,\ 7.4,\ 7.7,\ 8.4,\ 8.7,\ 8.9,\ 9.4$$

Determine her score in the 8th round if the upper quartile of all of her 8 scores is 8.85.

16 In training, a fighter pilot measures the number of seconds he blacks out over a number of flights. He constructs the following box and whisker plot for his data: As long as the pilot is not unconscious for more than 7 seconds, he will be safe to fly. The pilot concludes that he is safe to fly all the time. Is his conclusion correct? Explain your answer.



17 An advertising agency recorded the number of viewers within various age ranges of its newest television advertisement when it went to air. The results are shown in the table:

a Using the mean age of each age interval, find the five number summary.

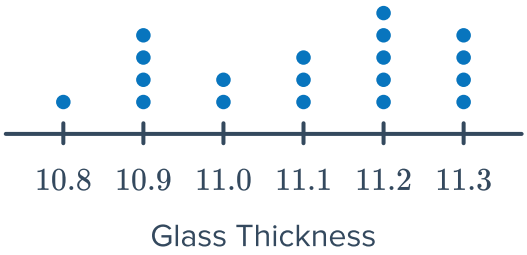
b According to the five number summary, approximately what percentage of viewers of the ad were aged between 28 and 48?

c The advertising agency was targeting viewers aged between 18 and 28. They would deem their ad as successful if at least 60% of viewers of the ad were in this age range.
According to the five number summary, were they successful in reaching their target viewers?

Age Interval	Frequency	Mean Age
$16-20$	350	18
$21-25$	150	23
$26-30$	200	28
$31-35$	300	33
$36-40$	300	38
$41-45$	300	43
$46-50$	400	48
$51-55$	300	53
$56-60$	200	58
$61-65$	50	63

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18 The glass windows for an airplane are cut to a certain thickness, but machine production means there is some variation. The thickness of each pane of glass produced is measured (in millimetres), and the dot plot shows the results:



Glass Thickness

a Find the median thickness, to two decimal places.

b Find the interquartile range.

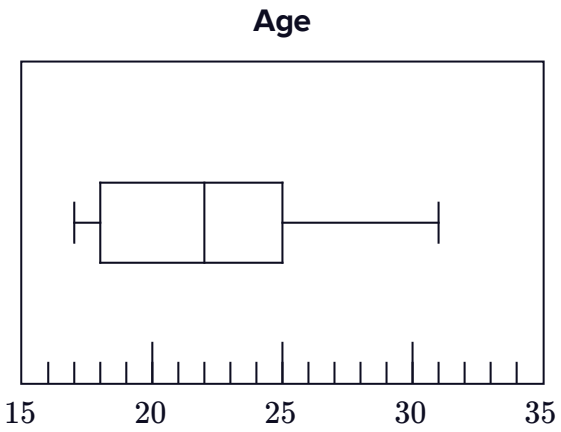
c Construct a box plot to represent the data.

d What percentage of thicknesses were between 10.8 mm and 11.2 mm inclusive? Round your answer to two decimal places.

e According to the box plot, in which quartile are the results the most spread out?

f Which statistics cannot be found from a box plot?

19 The box plot shows the age at which a group of people got their driving licences:



a What is the oldest age?

b What is the youngest age?

c What percentage of people were aged from 18 to 22?

d The middle 50% of responders were within how many years of one another?

e In which quartile are the ages least spread out?

f The bottom 50% of responders were within how many years of one another?

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